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# A Literature Survey on Smart Farming Systems for Minimum Support Price (MSP) Forecasting

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***Abstract*—Agriculture plays an important role in economic growth and farmer livelihood. However, many farmers still face difficulties in receiving fair prices for their crops because of market uncertainty, lack of pricing transparency, and dependence on intermediaries. Although the Minimum Support Price (MSP) system was introduced to protect farmers from price fluctuations, its practical benefits are often limited. This paper presents a literature survey on smart farming technologies and proposes a conceptual framework that combines Artificial Intelligence (AI) and Blockchain for transparent MSP-based agricultural trading. The AI component helps predict crop prices using historical and market-related data, while Blockchain provides secure and transparent transaction management through distributed ledgers and smart contracts. The proposed approach aims to support fair pricing, reduce dependency on middlemen, and improve trust between farmers and buyers. In addition, the study identifies research gaps and highlights future opportunities for integrating intelligent technologies in agriculture.**

***Index Terms*—Smart Agriculture, Artificial Intelligence, Blockchain, Minimum Support Price (MSP), Agricultural Trading, Smart Contracts.**

## I. Introduction

Agriculture plays an important role in economic development and food security, especially in countries like India where a large population depends on farming for livelihood. Even though agricultural production has improved over the years, farmers still face many challenges such as unstable crop prices, lack of market transparency, delayed payments, and dependence on intermediaries. In many situations, farmers are forced to sell their produce at lower prices due to limited market information and uncertain trading conditions. To address this issue, the Minimum Support Price (MSP) system was introduced to provide a minimum guaranteed price for selected crops. However, problems such as lack of awareness, limited procurement facilities, and delayed implementation reduce its effectiveness in practice.

In recent years, technologies such as Artificial Intelligence (AI), Blockchain, and Internet of Things (IoT) have shown promising applications in agriculture. AI techniques are widely used for crop prediction, weather forecasting, and agricultural price estimation, helping farmers make better decisions using historical and market-related data. Blockchain technology improves transparency and security by maintaining tamper-resistant transaction records and supporting smart contracts between stakeholders. IoT-based systems also help in monitoring soil conditions, temperature, and irrigation in real time.

Although these technologies provide useful solutions, most existing agricultural systems focus on individual tasks such as prediction, monitoring, or supply chain management. Very few studies combine intelligent crop price prediction with secure and transparent MSP-based trading. Therefore, there is a need for an integrated system that can support fair pricing and improve farmer confidence in agricultural markets.

This paper presents a literature survey and conceptual framework for an Artificial Intelligence and Blockchain-based MSP agricultural trading system. The study reviews existing agricultural technologies, identifies research gaps, and proposes an integrated framework to support transparent and secure agricultural transactions. The main contributions of this work include:

- Review of existing AI, Blockchain, and IoT-based agricultural systems.
- Comparative analysis of current agricultural technologies and their limitations.
- A conceptual framework integrating AI and Blockchain for MSP-based agricultural trading.
- Identification of research gaps and future research directions.

## II. Literature Survey

Recent progress in smart agriculture has increased the use of technologies such as Artificial Intelligence (AI), Blockchain, and Internet of Things (IoT) to solve farming related problems.


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These technologies mainly support crop prediction, farm monitoring, agricultural trading, and decision making. However, many existing systems focus only on individual agricultural tasks and do not provide a complete MSP oriented trading framework.

Artificial Intelligence has become useful in agriculture for improving prediction and planning. AI based farming systems help farmers make better decisions using historical and environmental data. Explainable AI approaches have improved transparency in agricultural recommendations and support farm planning [1]. Similarly, AI models have been used for agricultural demand prediction and crop price estimation to reduce uncertainty in farming decisions [5]. Machine learning techniques also support crop recommendation and price forecasting, helping farmers understand possible market trends [6]. Climate aware AI systems further assist in analyzing environmental changes affecting crop productivity [8]. Although these systems improve forecasting, most of them focus only on prediction and do not support secure agricultural trading or MSP based selling decisions.

Blockchain technology has also received attention in agriculture because of its ability to improve security and transparency. Several studies have shown that Blockchain can maintain secure records and improve traceability in agri food supply chains [4], [11]. Smart contracts have further improved trust between farmers and buyers by automating agricultural transactions [12]. Blockchain based agricultural systems also support secure data sharing and decentralized management [15]. In addition, Blockchain has been identified as a useful technology for sustainable and transparent agricultural supply chains [16], [17]. However, most Blockchain studies mainly focus on traceability and secure record keeping without considering MSP based crop pricing support.

IoT based smart farming systems improve agriculture through real time monitoring of soil, temperature, humidity, and irrigation conditions. Integrated Blockchain IoT frameworks have improved secure agricultural monitoring and reliable farm management [3]. Some studies also discussed combining IoT, AI, and Blockchain to create smart agricultural ecosystems [7]. However, most IoT systems mainly focus on crop monitoring rather than agricultural market transactions.

Digital agricultural marketplaces have been introduced to improve communication between farmers and buyers. Online platforms support direct trading and reduce intermediary involvement [2], [10]. Decentralized agricultural marketplaces have also improved transparency through distributed systems [9]. Some AI driven agricultural platforms help farmers by providing recommendations and market support [13], while fair pricing systems improve pricing awareness [14]. Even so, many existing platforms still lack intelligent price prediction and MSP focused support.

Based on the reviewed studies, it is observed that AI mainly supports prediction, Blockchain improves transparency, and IoT enables real time farm monitoring. However, most existing solutions work independently and do not combine intelligent prediction with secure MSP based agricultural trading. Therefore, there is a need for an integrated framework that supports price prediction, transparent transactions, and fair pricing for farmers.

## III. Comparative Analysis of Existing Studies

Existing agricultural technologies mainly focus on solving specific farming problems instead of providing a complete solution for transparent agricultural trading. Technologies such as Artificial Intelligence (AI), Machine Learning (ML), Blockchain, and Internet of Things (IoT) are widely used in agriculture for prediction, monitoring, and transaction management. However, their combined use for MSP oriented agricultural trading is still limited.

Artificial Intelligence and Machine Learning are mainly used for crop demand estimation, agricultural forecasting, and price prediction. AI based smart farming systems help farmers understand crop related decisions and improve prediction accuracy [1]. Demand prediction and crop forecasting models also support farmers in planning cultivation and marketing activities [5], [6]. Climate related prediction systems further help in understanding environmental effects on agriculture [8]. Even though these systems improve forecasting, they mainly focus on recommendations and lack support for secure transactions and MSP based selling guidance.

Blockchain technology improves transparency and trust in agriculture by maintaining secure records of transactions. Several studies show that Blockchain supports supply chain tracking and improves agricultural traceability [4], [11]. Smart contracts also help automate agreements between farmers and buyers and reduce intermediary involvement [12]. Blockchain based systems further improve secure data storage and transparency in agricultural activities [15], [16]. However, most Blockchain applications mainly focus on traceability and do not include intelligent crop price prediction or MSP related decision support.

IoT based agricultural systems support real time monitoring of farming conditions such as soil moisture, humidity, and temperature. Integrated IoT, AI, and Blockchain systems have shown better farm monitoring and secure data handling [3], [7]. However, these systems mainly focus on farm management and provide limited support for agricultural market transactions.

Digital agricultural marketplaces have improved farmer access to buyers and reduced intermediary dependency. Several platforms support direct crop selling and improve communication between farmers and buyers [2], [9], [10]. Some AI supported systems also help farmers with recommendations and pricing awareness [13], [14]. Still, many platforms lack intelligent forecasting, transparent pricing, and MSP based support.

Table I compares existing agricultural technologies based on objectives, limitations, and MSP support. Most current systems focus only on prediction, monitoring, or supply chain transparency. Very few studies combine intelligent crop price prediction with secure and transparent agricultural trading. Therefore, there is a need for an integrated framework that


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TABLE I
COMPARATIVE ANALYSIS OF EXISTING TECHNOLOGIES

| Technology | Objective | Limitation | MSP Support |
|---|---|---|---|
| Blockchain | Supply chain transparency | No price prediction | No |
| Machine Learning | Crop price prediction | No secure trading | No |
| ML + Blockchain | Smart agriculture | Limited market support | No |
| Blockchain + IoT | Farm monitoring | Limited pricing support | No |
| AI + Blockchain | MSP based trading | Conceptual framework | Yes |

combines Artificial Intelligence and Blockchain to support fair pricing, secure transactions, and MSP based agricultural trading.

## IV. PROBLEM STATEMENT

Agriculture is one of the major sources of livelihood, especially in countries such as India. However, farmers still face several challenges in receiving fair prices for their agricultural products. Market price fluctuations, lack of transparency, delayed procurement, and dependence on intermediaries often reduce farmer income. Due to financial pressure and limited market awareness, many farmers are forced to sell crops at lower prices, resulting in distress selling.

The Minimum Support Price (MSP) system was introduced to protect farmers by providing a guaranteed minimum price for selected crops. Even though MSP offers economic support, its implementation is often limited because of poor awareness, delayed payments, and restricted access to procurement facilities. As a result, many farmers do not fully benefit from MSP policies.

Several technologies such as Artificial Intelligence, Machine Learning, and Blockchain have been introduced to improve agricultural systems. AI based models support crop price prediction and agricultural planning using historical and environmental data. However, most systems mainly focus on prediction and are not connected with transparent agricultural trading. Similarly, Blockchain technology improves security and trust in transactions, but its use in MSP based agricultural trading remains limited.

Digital agricultural marketplaces have improved communication between farmers and buyers, but many platforms still face problems such as pricing uncertainty, lack of transparency, and delayed payments. Most existing solutions focus only on individual agricultural challenges and do not provide an integrated system for prediction, secure transactions, and MSP support.

Therefore, there is a need for a unified framework that combines Artificial Intelligence and Blockchain within an MSP based agricultural system. Such a framework can improve price prediction, increase transparency, reduce intermediary dependency, and help farmers make better selling decisions.

## V. PROPOSED METHODOLOGY

The proposed system introduces a smart agricultural trading framework that combines Artificial Intelligence (AI) and Blockchain technology to support secure and transparent MSP-based crop trading. The main objective is to help farmers make better selling decisions by predicting crop prices and reducing dependence on intermediaries. Unlike traditional systems that separately focus on prediction or transaction security, the proposed framework combines both functions in a single platform.

The framework consists of four main components: Data Collection Layer, Artificial Intelligence Module, Blockchain Module, and Smart Marketplace Interface. These components work together to improve transparency and efficiency in agricultural trading. Fig. 1 shows the overall workflow of the proposed system.

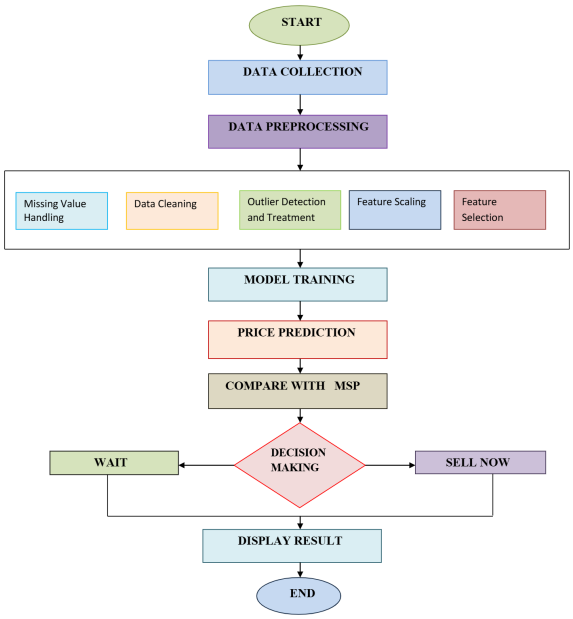



Fig. 1. Flow of data and interaction between AI, Blockchain, and the marketplace module.

### A. Data Collection Layer

The system collects agricultural information such as crop details, historical prices, rainfall, temperature, weather reports, and MSP records. Real-time market information from mandis and online agricultural platforms is also considered to improve prediction quality. Before model training, preprocessing techniques are applied to remove missing values and normalize the dataset.

### B. Artificial Intelligence Module

The AI module predicts expected crop prices using historical market trends and environmental conditions. Machine Learning techniques such as Random Forest, Gradient Boosting, and LSTM may be used to improve prediction accuracy. The predicted price is compared with MSP values to guide farmers in making better selling decisions and reducing distress selling.


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**Algorithm 1** AI and Blockchain-Based MSP Agricultural Trading Framework

**Require:** Crop Details, Historical Data, MSP Information
**Ensure:** Price Recommendation and Secure Transaction

```
Load agricultural and MSP dataset
Train AI model using historical crop and weather data
while System is active do
    Farmer enters crop information
    Predict crop price using AI model
    Compare predicted price with MSP
    if Predicted Price > MSP then
        Recommend waiting for better market conditions
    else
        Recommend selling at MSP
    end if
    if Buyer confirms purchase then
        Execute Blockchain smart contract
        Verify transaction details
        Record transaction securely
        Process payment
    end if
end while=0
```

### C. Blockchain Module

Blockchain technology is used to improve transparency and security in agricultural transactions. Once a farmer decides to sell, transaction details are securely stored in a decentralized ledger. Smart contracts automate agreements between farmers and buyers and ensure payment release only after predefined conditions are satisfied. This reduces intermediary dependency and improves trust.

### D. Smart Marketplace Interface

The marketplace interface allows farmers to upload crop information, view predicted prices, and compare them with MSP values. Buyers can access crop details and place bids accordingly. The interface is designed to be simple and accessible, with support for mobile devices and local languages to improve usability among rural farmers.

### E. Working Process

The process begins when the farmer enters crop details into the system. The AI module predicts future crop prices and compares them with MSP information. Based on the result, the system provides suitable selling recommendations. If the farmer agrees to sell, Blockchain smart contracts handle secure transaction verification and payment processing.

### F. Advantages of the Proposed System

The proposed framework improves price transparency, supports better farmer decision-making, and reduces dependence on intermediaries. AI improves crop price estimation, while Blockchain ensures secure and tamper-proof transactions. Overall, the system aims to create a fair and efficient agricultural trading environment.

## VI. Expected Outcomes and Discussion

The proposed AI and Blockchain-based agricultural trading framework is expected to improve transparency and fairness in MSP-based crop trading. By combining crop price prediction with secure transaction management, the system aims to help farmers receive better prices and reduce dependency on intermediaries.The Artificial Intelligence module is expected to assist farmers by predicting crop prices using historical records, weather conditions, rainfall, and MSP information. Based on the prediction, the system may recommend whether farmers should sell immediately or wait for better market conditions. This support can help reduce distress selling and improve decision-making.

The Blockchain module is expected to improve trust and security during transactions. Since all records are stored in a decentralized ledger, chances of data tampering may be reduced. Smart contracts can automate transactions and ensure payments are completed only after required conditions are verified, reducing delays and disputes.Compared to existing systems that mainly focus on prediction or monitoring, the proposed framework combines prediction, transparency, and MSP guidance within a single platform. However, practical issues such as poor internet access, limited digital literacy, and Blockchain implementation costs may affect adoption. With proper awareness and government support, the system may help improve fair agricultural trade and farmer income stability.

## VII. Research Gap and Future Scope

Existing studies show that Artificial Intelligence, Blockchain, and IoT technologies have improved smart agriculture in areas such as crop prediction, monitoring, and supply chain management. However, most systems focus on individual tasks and do not provide a complete MSP-based agricultural trading solution.A major research gap exists in integrating AI-based crop price prediction with Blockchain-supported secure transactions. Many AI systems mainly focus on prediction, while Blockchain applications are often limited to traceability and secure record storage. Similarly, digital marketplaces rarely provide MSP-aware recommendations or transparent pricing support for farmers.

Another limitation is the lack of focus on distress selling, delayed payments, and intermediary dependency. Many farmers still struggle to access reliable market information and fair pricing mechanisms.Future work can focus on developing real-time and scalable systems that combine AI prediction with Blockchain-based secure trading. Additional improvements may include IoT-based farm monitoring, satellite data integration, and multilingual mobile applications to improve accessibility for rural farmers. Real-world testing with farmers and government agencies may further improve system effectiveness and adoption.

## VIII. Conclusion & Future Work

This paper presented a literature survey and a conceptual framework for an Artificial Intelligence and Blockchain-based


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agricultural trading system aimed at improving transparency and fairness in MSP-based markets. From the reviewed studies, it was observed that most existing agricultural technologies mainly focus on separate areas such as crop prediction, monitoring, or supply chain tracking rather than providing a complete trading solution for farmers. The proposed framework combines Artificial Intelligence for crop price prediction and Blockchain for secure agricultural transactions. By analyzing historical prices, weather conditions, and MSP information, the system can help farmers make better selling decisions and reduce distress selling. At the same time, Blockchain technology improves transparency by securely recording transactions and supporting smart contract-based agreements between farmers and buyers. The proposed approach may reduce dependency on intermediaries, improve trust among stakeholders, and support fair agricultural pricing. Although the framework is conceptual, it highlights the possibility of building a more transparent and efficient agricultural trading system. In future work, the proposed model can be implemented using real agricultural datasets to evaluate prediction accuracy and system performance. Additional improvements such as IoT-based monitoring, satellite data integration, and multilingual mobile applications may further improve accessibility and adoption among rural farming communities.

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